Superlinear Quantum Query Lower Bounds for Every 2-Connected Subgraph, and Better Bounds for Complete Bipartite Graphs and Small Cliques

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Abstract

For a fixed graph H, consider the problem of deciding whether an N-vertex graph, given by its adjacency matrix, contains H as a subgraph. We show that for every fixed 2-connected graph H on v vertices, every bounded-error quantum algorithm makes $\Omega(N^{1+1/(6(v-1))})$ queries. To our knowledge, this gives the first superlinear lower bounds for the four-cycle, $\Omega(N^{19/18})$, for every even cycle, for $K_{r,r}$ with $r \leq 9$ and for every unbalanced $K_{a,b}$. A reduction to a single block extends the bounds to many connected graphs with cut vertices, for example two four-cycles sharing a vertex. For triangles the bound is the $\Omega(N^{13/12})$ of Cornelissen, Gilani and Zhou, whose method we follow. Counting the extensions of larger patterns gives more for dense H: $\Omega(N^{7/6-\theta/6})$ with $\theta = (3a+3b-8)/(3(a-1)b+1)$ for $K_{a,b}$, $3 \leq a \leq b$, which is $N^{41/38}$ for $K_{3,3}$ and $N^{615/542}$ for $K_{10,10}$, where the bound above gives $N^{115/114}$; for $K_{r,r}$ with $3 \leq r \leq 18$ this is the best bound known to us. We also obtain $\Omega(N^{16/15})$ for $K_{2,3}$, $\Omega(N^{15/14})$ for $K_{2,b}$ with $b \geq 4$ and $\Omega(N^{11/10})$ for the octahedron. For cliques the method gives $\Omega(N^{7/6-1/(3(k+1))})$ for K_k with $4 \leq k \leq 7$, which is $N^{11/10}$, $N^{10/9}$ and $N^{47/42}$ for K_4, K_5 and K_6, improving the previous exponents 13/12, 13/12 and 11/10. Through a reduction of Gilani and Zhou this gives the same bounds for connected graphs of chromatic number 4, 5 and at least 6. The proofs compare the computation on a planted copy with the computation on random graphs conditioned to contain no copy. We state this comparison once for blow-ups of H, with three hypotheses on the conditioned random graph. For 2-connected H the hypotheses follow from degree bounds. For cliques we need a version of the projection-summing lemma of Cornelissen, Gilani and Zhou without a spectral gap, and polynomial concentration of extension counts.

1 Introduction

In the adjacency matrix model, an algorithm learns about an unknown N-vertex graph by asking whether two vertices are adjacent. For a fixed graph H, the problem Sub_H asks whether the input contains a subgraph isomorphic to H. Learning graphs solve Sub_H with $O(N^{2-2/v-\alpha_H})$ queries for every H on $v \geq 3$ vertices, where $\alpha_H > 0$ [8, 11], and every fixed path can be detected with $O(N)$ queries [2]. Unstructured search gives $\Omega(N)$ for every H with an edge. Childs and Kothari asked whether detecting any fixed subgraph needs superlinearly many queries [3, Section 5]. The certificate complexity barrier [9, 10] limits the nonnegative adversary method to $O(N)$ for all these problems.

Gilani and Zhou [6] gave the first superlinear lower bounds: for K_k with $k \geq 4$, with exponents 19/18, 13/12, 11/10 and 9/8 for $k = 4, 5, 6, 7$; for every connected graph of chromatic number at least four; and for the complete bipartite graphs $K_{r,r}$ with $r \geq 10$. Cornelissen, Gilani and Zhou [5] then proved $\Omega(N^{13/12})$ for triangles, and so for every connected non-bipartite graph, with

a comparison method built on Belovs' approach to k-distinctness [1]. Gilani and Zhou list the four-cycle and the graphs $K_{r,r}$ with $r < 10$ as open, and ask more generally which fixed subgraphs can be detected with $O(N)$ queries [6, Section 1.4].

Results. Our first result settles the four-cycle and, more generally, every 2-connected pattern.

Theorem 1.1. *Let H be a fixed 2-connected graph on $v \geq 3$ vertices. Every bounded-error quantum algorithm that detects, or finds, a copy of H in an N-vertex graph in the adjacency matrix model makes $\Omega(N^{1+1/(6(v-1))})$ queries.*

A block of a graph is a maximal 2-connected subgraph or a bridge. Attaching known gadgets to every vertex reduces the detection of a suitable block of H to the detection of H (Lemma 4.1), so Theorem 1.1 extends as follows.

Corollary 1.2. *Let H be a fixed connected graph, and let B be a block of H with $v_B \geq 3$ vertices. Suppose that B has the most edges among the blocks of H and at most one other block of H is isomorphic to B, or that every component of $H - E(B)$ has fewer than $e(H)/2$ edges. Then detecting or finding H takes $\Omega(N^{1+1/(6(v_B-1))})$ quantum queries.*

The first condition holds whenever the 2-core of H is 2-connected, for example for a cycle with trees attached, and also for two four-cycles that share a vertex or are joined by a path, which gives $\Omega(N^{19/18})$. The second holds, for example, for a chain of three four-cycles in which the outer ones are attached at different vertices of the middle one. For triangles, Theorem 1.1 is the bound of [5]. For non-bipartite H the bound $N^{13/12}$ of [5] is at least as large, so the new cases are the bipartite ones (Table 1).

H	best previous bound known to us	Theorem 1.1
$C_4 = K_{2,2}$	$\Omega(N)$	$N^{19/18}$
$C_{2\ell}$, $\ell \geq 2$	$\Omega(N)$	$N^{1+1/(6(2\ell-1))}$
$K_{a,b}$, $2 \leq a \leq b$	$N^{\beta_r - o(1)}$ for $a = b = r \geq 10$ [6]	$N^{1+1/(6(a+b-1))}$

Table 1: Bipartite 2-connected patterns. For $K_{a,b}$ with $a < b$ we know of no earlier superlinear bound. For $K_{10,10}$ Theorem 1.1 gives $N^{115/114}$, above the $N^{181/180-o(1)}$ of [6]; for $r \geq 11$ the bounds of [6] are larger. Corollary 6.3 improves Theorem 1.1 for $K_{a,b}$ with $a + b \geq 5$.

For cliques the same comparison gives more, because the hard inputs can be denser.

Theorem 1.3. *For every $4 \leq k \leq 7$, detecting or finding K_k takes $\Omega(N^{7/6-1/(3(k+1))})$ quantum queries.*

For K_7, Theorem 1.3 only removes the $o(1)$ from the exponent of [6]. Gilani and Zhou asked for any improvement over 19/18 for K_4 [6, Section 1.4]. The triangle bound of [5] already gives 13/12; Theorem 1.3 gives 11/10. By [6, Theorem 1.3(i)], which reduces K_r detection with $r = \min\{\chi, 6\}$ to the detection of any connected graph of chromatic number χ, we obtain the following.

Corollary 1.4. *Let H be a fixed connected graph of chromatic number χ. Detecting or finding H takes $\Omega(N^{11/10})$ quantum queries if $\chi = 4$, $\Omega(N^{10/9})$ if $\chi = 5$, and $\Omega(N^{47/42})$ if $\chi \geq 6$.*

For $\chi = 4, 5$ this improves 13/12 from [5]. For $\chi \geq 6$ it improves 11/10 from [6], except where the bounds of [6, Theorem 1.3(ii)] for large chromatic number are stronger.

Section 6 replaces the degree bounds by extension counts. This gives a criterion for every connected H (Theorem 6.2) and the following bounds.

k	previous lower bound	Theorem 1.3	upper bound [8, 11]
4	$N^{13/12}$ [5]; $N^{19/18-o(1)}$ [6]	$N^{11/10}$	$N^{3/2-g(4)}$
5	$N^{13/12}$ [5]; $N^{13/12-o(1)}$ [6]	$N^{10/9}$	$N^{8/5-g(5)}$
6	$N^{11/10-o(1)}$ [6]	$N^{47/42}$	$N^{5/3-g(6)}$
7	$N^{9/8-o(1)}$ [6]	$N^{9/8}$	$N^{12/7-g(7)}$

Table 2: Exponents for K_k detection. Here $g(k) = \Theta(k^{-3}) > 0$, as stated in [6]. The bounds attributed to [5] follow from the triangle bound by adding universal vertices.

H	best previous bound known to us	Corollary 6.3, Remark 6.4
$K_{2,3}$	$N^{25/24}$ (Theorem 1.1)	$N^{16/15}$
$K_{2,b}$, $b \geq 4$	$N^{1+1/(6(b+1))}$ (Theorem 1.1)	$N^{15/14}$
$K_{3,3}$	$N^{31/30}$ (Theorem 1.1)	$N^{41/38}$
$K_{10,10}$	$N^{115/114}$ (Theorem 1.1)	$N^{615/542}$
$K_{a,b}$, $3 \leq a \leq b$	$N^{1+1/(6(a+b-1))}$, or [6] for $a = b \geq 11$	$N^{7/6-\theta_{a,b}/6}$
$K_{2,2,2}$	$N^{13/12}$ [5]	$N^{11/10}$

Table 3: Bounds from extension counts, with $\theta_{a,b} = (3a+3b-8)/(3(a-1)b+1)$. For $K_{r,r}$ the new exponent is larger than that of [6] exactly for $3 \leq r \leq 18$; $615/542 \approx 1.135$.

The method. We follow [5]. The algorithm knows a host graph G, the blow-up of H in which every vertex becomes a part of size n and every edge a complete bipartite graph. Negative inputs are Bernoulli(p) subgraphs of G conditioned to contain no copy of H. Positive inputs plant a uniformly random copy and condition on it being the only one. A comparison map sends the workspace on negative inputs to the workspace on positive inputs. It uses a few negative inputs per candidate and an averaging operator built from conditional resampling, and its error per query determines the bound. We state this comparison for every H in Proposition 3.1, under three hypotheses on the conditioned distribution: planting rarely creates another copy; inside the star of a vertex, each edge interacts with few others; and each edge interacts with few edges of the whole input. With averaging parameter $R = (np)^{2/3}$, whenever the hypotheses hold with bounded pd_s/R and negligible failure probability, the bound is $t = \Omega(n^{7/6}p^{1/6})$, whatever H is. So larger admissible densities give larger bounds.

For a 2-connected H, every interaction is witnessed by a spanning forest of a copy whose $v-2$ edges are present in the input. So all three hypotheses reduce to degree bounds, and they hold up to $p \approx n^{-(v-2)/(v-1)}$. For triangles this is the density $n^{-1/2}$ of [5]. For K_k the planting condition allows $p \approx n^{-2/(k+1)}$, but the star and interaction hypotheses then fail in the form used by [5]. We prove a version of their projection-summing lemma that needs no spectral gap (Lemma 3.2), which handles stars for $k \leq 7$. We bound the interactions by clique extension counts, using the polynomial concentration inequality of Kim and Vu [7] (Lemma 5.1). Section 6 applies the same inequality to the star hypothesis as well, which for $K_{r,r}$ allows $p \approx n^{-(6r-8)/(3r^2-3r+1)}$.

Which subgraphs are linear? Paths, stars with two subdivided legs and subdivided claws can be detected with $O(N)$ queries [2]. By [5, 6], every connected non-bipartite graph is superlinear, and by Theorem 1.1 and Corollary 1.2 so are many bipartite graphs, including all connected graphs

whose 2-core is 2-connected. Lemma 4.2 adds every graph with a 2-connected block through a vertex u at which every component of $H - u$ has at most $e(H)/2$ edges, for example three or more four-cycles sharing a vertex. We conjecture that Lemma 4.1 holds for every connected graph and every block with the most edges. A computer search over 337 small graphs, 200 of them with several largest blocks of the same shape, found no counterexample. The conjecture would make every connected graph with a cycle superlinear and leave only trees open.

Organization. Section 2 recalls the framework of [5]. Section 3 proves Proposition 3.1. Section 4 proves Theorem 1.1 and Corollary 1.2. Section 5 proves Theorem 1.3. Section 6 proves Theorem 6.2, Corollary 6.3 and the octahedron bound of Remark 6.4. Section 7 discusses the limits of the method.

2 The comparison framework

We use the setting of [5, Sections 3 and 4]. Let J be a finite set of input bits. For a distribution ρ on $\{0,1\}^J$, let $\mathcal{H}_\rho$ be the span of $\{|x\rangle : x \in \operatorname{supp}\rho\}$ and $|\rho\rangle := \sum_x \sqrt{\rho(x)}|x\rangle$. For a finite-dimensional workspace W, a joint vector is $f = \sum_x \sqrt{\rho(x)} f^x \otimes |x\rangle \in W \otimes \mathcal{H}_\rho$, with $\|f\|^2 = \mathbb{E}_{x\sim\rho}\|f^x\|^2$ and $\|f\|_\infty = \max_{x\in\operatorname{supp}\rho}\|f^x\|$. For $e \in J$, the phase operator Q_e maps $|x\rangle$ to $(-1)^{x_e}|x\rangle$. Operators on the input register act as the identity on W.

Theorem 2.1 ([5, Theorem 4.1])**.** *Let $\rho^\pm$ be distributions on inputs, with phase operators $Q_e^\pm$, and let Υ be a linear map from $\mathcal{H}_{\rho^-}$ into a space $\mathcal{H}^+$ on which Q_e^+ acts, with $\Upsilon|\rho^-\rangle = |\rho^+\rangle$. Suppose that for every $e \in J$, every workspace and every f, $\|(Q_e^+\Upsilon - \Upsilon Q_e^-)f\|^2 \le \alpha_e\|f\|^2 + \beta_e\|f\|_\infty^2$. If a correct output can be verified exactly with $\bar{s}$ queries by a verifier that always rejects on negative inputs, then every t-query algorithm with average success probability σ under ρ^+ satisfies $\sqrt{\sigma} \le (t+\bar{s})(\max_e \alpha_e + \sum_e \beta_e)^{1/2}$.*

In [5], $\mathcal{H}^+$ carries a label register in addition to the input; we use the same construction in Section 3. Queries to pairs that are known to be absent act as the identity on both sides, so they commute with Υ and add no error.

Conditioned Bernoulli inputs. Let $\mathcal{C}$ be a family of nonempty subsets of J, let $\Omega := \{x : x_C \ne 1_C \text{ for all } C \in \mathcal{C}\}$, $\nu := \text{Bernoulli}(p)^{\otimes J}$ with $0 < p \le 1/2$, and $\mu := \nu(\,\cdot \mid \Omega)$. Since Ω is downward closed, after fixing all coordinates but j, either x_j is forced to 0 or it is Bernoulli(p). The conditional average E_j replaces the workspace vector by its conditional mean over x_j, and $D_j := \mathbb{1} - E_j$. Both are orthogonal projections, and $\|D_j f\|^2$ is the conditional variance at j. For weights $\omega_j \ge 0$, $W := \sum_j \omega_j D_j$ and $R > 0$, the averaging operator is $F := (\mathbb{1} + W/R)^{-1}$. Let $K_e := [D_e, Q_e]$.

Lemma 2.2 ([5, Lemma 4.2, (4.14), (4.15)])**.** *F is self-adjoint, $0 \preceq F^2 \preceq F \preceq \mathbb{1}$, $F|\mu\rangle = |\mu\rangle$, and $\|Ff\|_\infty \le \|f\|_\infty$. If $W' \succeq W$ and $F' := (\mathbb{1} + W'/R)^{-1}$, then $(F')^2 \preceq F' \preceq F$. For every operator X on the input register, $[X,F] = R^{-1}F[W,X]F$, and $\langle Ff, WFf\rangle \le \frac{R}{4}\|f\|^2$. Moreover $\|K_e\| \le 2\sqrt{p}$, $\|K_e f\|_\infty \le 2\|f\|_\infty$, $[D_j, Q_e] = 0$ for $j \ne e$, and $[Q_e, F] = \frac{\omega_e}{R} F K_e F$.*

For distinct i, j, let $A_{ij}(x)$ be the indicator that some $C \in \mathcal{C}$ contains i and j and has $x_c = 1$ for all $c \in C \setminus \{i,j\}$. It depends only on the coordinates outside $\{i,j\}$, and A_{ij} also denotes multiplication by it. For $B \subseteq J$ and $i \in B$, put $N_{i,B}(x) := \sum_{j\in B\setminus\{i\}} A_{ij}(x)$.

Lemma 2.3 ([5, Lemma 4.3, (5.33)])**.** *For distinct i, j: $\|[D_i, D_j]f\|^2 \le p^2(1+p)(\|A_{ij}D_i f\|^2 + \|A_{ij}D_j f\|^2)$; $\|([D_i,D_j]f)^x\| \le 2pA_{ij}(x)\|f\|_\infty$; and $\|[D_i, K_e]\| \le 2p$ for $i \ne e$.*

Lemma 2.4 ([5, Lemma 4.5]). *Let $B \subseteq J$, $L_B := \sum_{i\in B} D_i$, $F_B := (\mathbb{1} + L_B/R)^{-1}$, and let $\mathcal{G}_B$ be an event determined by the coordinates outside B with $\mu(\mathcal{G}_B^c) \le b$. If $N_{i,B}(x) \le d$ for all $i \in B$ and $x \in \operatorname{supp}\mu \cap \mathcal{G}_B$, and $pd \le 1/10$, then $\sum_{i\in B}\langle D_i f, F_B D_i f\rangle \le 3R\|f\|^2 + |B|b\|f\|_\infty^2$.*

Lemma 2.5 ([5, Lemma 4.6]). *Let $W = \sum_j \omega_j D_j$ with $\sum_j \omega_j \le \omega$, fix i, and let $\mathcal{G}$ be an event, possibly depending on all coordinates, with $\mu(\mathcal{G}^c) \le b$ and $\sum_{j\ne i}\omega_j A_{ij}(x) \le d$ for $x \in \mathcal{G}$, where $0 \le d \le \omega$ and $pd \le 2/5$. For $R \ge 1$ and $F = (\mathbb{1} + W/R)^{-1}$, $\|D_i F f\|^2 \le 4\|F D_i f\|^2 + \frac{p}{R}\|f\|^2 + \frac{64p^2\omega^2 b}{R^2}\|f\|_\infty^2$.*

Lemma 2.6 ([5, Lemma 5.2]). *Every increasing event has probability under μ at most its probability under ν.*

3 A comparison bound for blow-ups

Fix a graph H with $v \ge 3$ vertices, $e_H \ge 2$ edges and no isolated vertices. For $n \ge 1$, the blow-up $G = G_H(n)$ has vertex set $V(H) \times [n]$, with (a,x) and (b,y) adjacent when $ab \in E(H)$. The algorithm knows G; the input is a subgraph of G, and all other pairs are known to be absent. Let $J := E(G)$, let $\mathcal{F}$ be the family of edge sets of all subgraphs of G isomorphic to H, and $M := |\mathcal{F}|$. Some members of $\mathcal{F}$ may use several vertices of one part. Every vertex of G has degree at most $\Delta := (v-1)n$, $|J| \le e_H n^2$, and $M \ge n^v/|\mathrm{Aut}(H)|$, counting the copies with one vertex in each part. For every set X of $r \le v$ vertices of G, at most $C_H n^{v-r}$ members of $\mathcal{F}$ have vertex sets containing X, where $C_H := v^v$.

The negative distribution is $\mu^- := \nu(\,\cdot \mid \Omega^-)$, where Ω^- is the set of subgraphs of G containing no member of $\mathcal{F}$; all conditional operators below refer to μ^-. For $T \in \mathcal{F}$, let $\mathcal{G}_T$ be the event on the edges outside T that adding the edges of T creates no member of $\mathcal{F}$ other than T. The distribution μ_T samples the edges outside T from ν conditioned on $\mathcal{G}_T$ and sets the edges of T to one, and $\mu^+ := M^{-1}\sum_T \mu_T$, with T kept in a label register as in [5, (5.3)]. So $\mathcal{H}^+$ is spanned by $|T,y\rangle$ for $y \in \operatorname{supp}\mu_T$, a positive joint vector is $g = M^{-1/2}\sum_T\sum_y \sqrt{\mu_T(y)}\, g^{T,y} \otimes |T,y\rangle$, and $\|g\|^2 = M^{-1}\sum_T \|g^T\|^2$ with $g^T := \sum_y \sqrt{\mu_T(y)} g^{T,y} \otimes |y\rangle$. The phase Q_e^+ acts on y only.

For a vertex w of G, let $\mathrm{st}(w)$ be the set of edges of G at w.

Proposition 3.1. *Let $0 < p \le 1/2$ with $R := (np)^{2/3} \ge 1$, and let $0 \le b \le 1$ and $d_s, d_m \ge 0$. Assume:*

- *(P) $\mu^-(\mathcal{G}_T) \ge 1/2$ for every $T \in \mathcal{F}$.*
- *(S) For every vertex w there is an event $\mathcal{G}_w$ determined by the edges outside $\mathrm{st}(w)$, with $\mu^-(\mathcal{G}_w^c) \le b$, such that $N_{i,\mathrm{st}(w)}(x) \le d_s$ for every $i \in \mathrm{st}(w)$ and $x \in \operatorname{supp}\mu^- \cap \mathcal{G}_w$.*
- *(I) There is an event $\mathcal{G}_I$ with $\mu^-(\mathcal{G}_I^c) \le b$ such that $\#\{j \ne i : A_{ij}(x) = 1\} \le d_m$ for every edge i and $x \in \mathcal{G}_I$, and $pd_m \le 1/5$.*

Put $S := 2 + (pd_s)^2/R^2$. Every t-query algorithm that outputs the planted copy with average probability σ under μ^+ satisfies

$$\sqrt{\sigma} \le (t + e_H)\big(C_1 S\, n^{-7/3}p^{-1/3} + C_1 n^4 b\big)^{1/2},$$

where C_1 depends only on H.

When $pd_s \le R$, $S \le 3$. We first prove two lemmas.

Lemma 3.2. *Let $(\Pi_i)_{i\in I}$ be a finite family of orthogonal projections on a finite-dimensional Hilbert space and $L := \sum_i \Pi_i$. Suppose $\sum_i \|[\Pi_i, L]g\|^2 \le \kappa^2 \langle g, Lg\rangle$ for every g. For every $R > 0$, $F := (\mathbb{1} + L/R)^{-1}$ satisfies $\sum_i \langle \Pi_i f, F\Pi_i f\rangle \le (2 + \kappa^2/(4R^2))R\|f\|^2$.*

Proof. Let Y be the left side. The facts $0 \preceq F^2 \preceq F \preceq \mathbb{1}$, $LF = R(\mathbb{1} - F)$, $\langle Ff, LFf\rangle \le \frac{R}{4}\|f\|^2$ and $[X, F] = R^{-1}F[L, X]F$ are spectral identities that hold for every positive semidefinite L, as in Lemma 2.2. Writing $F\Pi_i = \Pi_i F + [F, \Pi_i]$ and $[F, \Pi_i] = R^{-1}F[\Pi_i, L]F$ gives, as in [5, (4.23)],

$$Y = \langle f, LFf\rangle + \frac{1}{R}\,\mathrm{Re}\sum_i \langle F\Pi_i f, [\Pi_i, L]Ff\rangle.$$

Since $LF = R(\mathbb{1} - F)$ and $0 \preceq F \preceq \mathbb{1}$, the first term is at most $R\|f\|^2$. Since $F^2 \preceq F$, $\sum_i \|F\Pi_i f\|^2 \le Y$. Also $\langle Ff, LFf\rangle \le \frac{R}{4}\|f\|^2$. By Cauchy–Schwarz and the hypothesis with $g = Ff$,

$$Y \le R\|f\|^2 + \frac{1}{R}\,Y^{1/2}\kappa\,\langle Ff, LFf\rangle^{1/2} \le R\|f\|^2 + \frac{\kappa}{2\sqrt{R}}\,Y^{1/2}\|f\|.$$

For $f \neq 0$, $y := (Y/(R\|f\|^2))^{1/2}$ satisfies $y^2 \le 1 + \frac{\kappa}{2R}y$, so $y \le \frac{\kappa}{4R} + (1 + \frac{\kappa^2}{16R^2})^{1/2}$ and $y^2 \le 2 + \frac{\kappa^2}{4R^2}$. □

Lemma 3.2 replaces the hypothesis $L^2 \succeq \gamma L$ of [5, Lemma 4.4] by the assumption that κ is not much larger than R.

Lemma 3.3. *In the setting of Lemma 2.4, but without the assumption $pd \le 1/10$,*

$$\sum_{i\in B}\langle D_i f, F_B D_i f\rangle \le \Big(2 + \frac{(pd)^2}{R^2}\Big)R\|f\|^2 + |B|b\|f\|_\infty^2.$$

Proof. Fix the coordinates outside B. The conditional distribution on B is again Bernoulli(p) conditioned on a downward-closed set, so Lemma 2.3 applies to it, and L_B and F_B act within it. Suppose the outside assignment lies in $\mathcal{G}_B$. For every x, at most d of the vectors $([D_i, D_j]g)^x$ with $j \in B \setminus \{i\}$ are nonzero. By Cauchy–Schwarz and Lemma 2.3, in the conditional norm,

$$\begin{aligned}\sum_{i\in B}\|[D_i, L_B]g\|^2 &\le d\,p^2(1+p)\sum_{i\in B}\sum_{j\in B\setminus\{i\}}\big(\|A_{ij}D_i g\|^2 + \|A_{ij}D_j g\|^2\big)\\ &= 2dp^2(1+p)\sum_{i\in B}\mathbb{E}\big[N_{i,B}(x)\|(D_i g)^x\|^2\big],\end{aligned}$$

using $A_{ij} = A_{ji}$ and $A_{ij}^2 = A_{ij}$. This is at most $2(1+p)(pd)^2\langle g, L_B g\rangle$. Lemma 3.2 with $\kappa^2 = 2(1+p)(pd)^2 \le 3(pd)^2$ bounds the conditional sum by $(2 + (pd)^2/R^2)R$ times the conditional $\|f\|^2$. If the outside assignment is not in $\mathcal{G}_B$, the conditional sum is at most $|B|$ times the conditional $\|f\|_\infty^2$, since each D_i is a projection and $F_B \preceq \mathbb{1}$. Averaging over the outside coordinates proves the claim. □

Proof of Proposition 3.1. Throughout, c denotes constants that depend only on H and may change from line to line, and $s := e_H$.

Local comparison. Put $Z := \nu(\Omega^-)$, $\nu_{\bar{T}} := \text{Bernoulli}(p)^{\otimes(J\setminus T)}$ and $Z_T := \nu_{\bar{T}}(\mathcal{G}_T)$. For $z \in \mathcal{G}_T$, the inputs that agree with z outside T and are negative are exactly those in which a proper subset of T is present: removing edges creates no copy, and the only copy in $(z, 1_T)$ is T. Hence $\mu^-(\mathcal{G}_T) = (1 - p^s)Z_T/Z$, and by (P),

$$r_T := \frac{Z}{(1-p)^s Z_T} = \frac{1-p^s}{(1-p)^s\mu^-(\mathcal{G}_T)} \le 2^{s+1}, \qquad \mu_T(z, 1_T) = \frac{\nu_{\bar{T}}(z)}{Z_T} = r_T\,\mu^-(z, 0_T).$$

For $y = (z, 1_T) \in \operatorname{supp}\mu_T$ and $U \subseteq T$, let $y^U := (z, 1_U)$. For a negative joint vector f, put $f_0^{(T)} := \sum_y \sqrt{\mu_T(y)} f^{y^\emptyset} \otimes |y\rangle$ and $f_i^{(T)} := \sum_y \sqrt{\mu_T(y)} f^{y^{\{i\}}} \otimes |y\rangle$ for $i \in T$, and define

$$\mathcal{T}_T f := \sum_{i\in T} f_i^{(T)} - (s-1) f_0^{(T)} = f_0^{(T)} + \sum_{i\in T} \big(f_i^{(T)} - f_0^{(T)}\big).$$

The coefficients sum to one, so $\mathcal{T}_T|\mu^-\rangle = |\mu_T\rangle$. The identity $\mu_T(z, 1_T) = r_T \mu^-(z, 0_T)$ gives $\|f_0^{(T)}\|^2 \le r_T \|f\|^2$. For $i \in T$ and $z \in \mathcal{G}_T$, the inputs $(z, 0_T)$ and $(z, 1_{\{i\}})$ are negative and differ only at i. Their total probability is $(1-p)^{s-1}\nu_{\bar{T}}(z)/Z$ and the conditional variance factor is $p(1-p)$, so they contribute $p(1-p)^s \nu_{\bar{T}}(z) Z^{-1} \|f^{(z,1_{\{i\}})} - f^{(z,0_T)}\|^2$ to $\|D_i f\|^2$. Distinct z give disjoint pairs, so $\|f_i^{(T)} - f_0^{(T)}\|^2 \le (r_T/p)\|D_i f\|^2$. Consequently

$$\|\mathcal{T}_T f\|^2 \le c\Big(\|f\|^2 + \frac{1}{p}\sum_{i\in T} \|D_i f\|^2\Big), \qquad \|\mathcal{T}_T f\|^2 \le \frac{c}{p}\|f\|^2. \tag{1}$$

For $e \notin T$, $Q_e^+ \mathcal{T}_T = \mathcal{T}_T Q_e^-$, since all inputs involved agree at e. For $e \in T$, the phase is -1 on y and $y^{\{e\}}$ and $+1$ on the other inputs used, so $(Q_e^+ \mathcal{T}_T - \mathcal{T}_T Q_e^-) f = -2\sum_{i\in T\setminus\{e\}} (f_i^{(T)} - f_0^{(T)})$, and

$$\|(Q_e^+ \mathcal{T}_T - \mathcal{T}_T Q_e^-) f\|^2 \le \frac{4 s r_T}{p} \sum_{i\in T\setminus\{e\}} \|D_i f\|^2. \tag{2}$$

Averaging. For a vertex w, put $L_w := \sum_{i\in \mathrm{st}(w)} D_i$ and $F_w := (\mathbb{1} + L_w/R)^{-1}$. For $T \in \mathcal{F}$, put $W_T := \sum_{w\in V(T)} L_w = \sum_i h_T(i) D_i$, where $h_T(i) := |i \cap V(T)| \le 2$ and $\sum_i h_T(i) \le v\Delta$, and $F_T := (\mathbb{1} + W_T/R)^{-1}$. By Lemma 2.2, $F_T^2 \preceq F_T \preceq F_w$ for $w \in V(T)$. Lemma 3.3 with $B = \mathrm{st}(w)$ and (S) give

$$\sum_{i\in \mathrm{st}(w)} \langle D_i f, F_w D_i f\rangle \le SR\|f\|^2 + \Delta b \|f\|_\infty^2. \tag{3}$$

Apply Lemma 2.5 with $W = W_T$, $\omega := v\Delta$, the event $\mathcal{G}_I$, and $d := \min\{2d_m, v\Delta\}$. Since $h_T \le 2$, $\sum_{j\ne i} h_T(j) A_{ij}(x) \le d$ on $\mathcal{G}_I$, and $pd \le 2/5$ by (I). With $\tau^2 := 16 p^2 (v\Delta)^2 b$, for every T and every edge i,

$$\|D_i F_T f\|^2 \le 4\|F_T D_i f\|^2 + \frac{p}{R}\|f\|^2 + \frac{4\tau^2}{R^2}\|f\|_\infty^2. \tag{4}$$

Sums over candidates. Fix a queried edge $e = uu'$, and let $A_e := \{T : e \in T\}$ and $B_e := \{T : e \notin T,\ V(T) \cap \{u, u'\} \ne \emptyset\}$. Then $|A_e| \le c n^{v-2}$ and $|B_e| \le c n^{v-1}$. For $i \in T$ and an endpoint w of i, $\|F_T D_i f\|^2 = \langle D_i f, F_T^2 D_i f\rangle \le \langle D_i f, F_w D_i f\rangle$. In the sum over $T \in A_e$ and $i \in T \setminus \{e\}$, charge i to the star at u or u' if it has one of them as an endpoint, and otherwise to the star at its first endpoint in a fixed order. An edge at u or u' lies in at most $c n^{v-3}$ members of A_e. Any other edge lies in at most $c n^{v-4}$ of them, and these edges are charged to at most vn stars. By (3) and (4),

$$\sum_{T\in A_e}\sum_{i\in T\setminus\{e\}} \|D_i F_T f\|^2 \le c\Big(n^{v-3} SR + \frac{n^{v-2} p}{R}\Big)\|f\|^2 + c\Big(n^{v-3}\Delta b + \frac{n^{v-2}\tau^2}{R^2}\Big)\|f\|_\infty^2. \tag{5}$$

In the sum over $T \in B_e$ and $i \in T$, an edge at u or u' lies in at most $c n^{v-2}$ members and is charged to the star at that vertex. Any other edge lies in at most $c n^{v-3}$ members and is charged to the star at its first endpoint, one of at most vn stars. So

$$\sum_{T\in B_e}\sum_{i\in T} \|D_i F_T f\|^2 \le c\Big(n^{v-2} SR + \frac{n^{v-1} p}{R}\Big)\|f\|^2 + c\Big(n^{v-2}\Delta b + \frac{n^{v-1}\tau^2}{R^2}\Big)\|f\|_\infty^2. \tag{6}$$

Query error. Define $\Upsilon f := M^{-1/2}\sum_T \sum_y \sqrt{\mu_T(y)}\,(\mathcal{T}_T F_T f)^y \otimes |T, y\rangle$, so that its component for T is $\mathcal{T}_T F_T f$, and $\Upsilon|\mu^-\rangle = |\mu^+\rangle$ because $F_T|\mu^-\rangle = |\mu^-\rangle$. The squared error at e is $M^{-1}\sum_T \|g_T\|^2$, where

$$g_T := (Q_e^+ \mathcal{T}_T - \mathcal{T}_T Q_e^-) F_T f + \mathcal{T}_T [Q_e^-, F_T] f.$$

If T contains neither u nor u', then $e \notin T$ and $h_T(e) = 0$, so $g_T = 0$. Otherwise, by Lemma 2.2, $[Q_e^-, F_T] = \frac{h_T(e)}{R} F_T K_e F_T$, of norm at most $4\sqrt{p}/R$.

For $T \in A_e$, (2) and (5) bound the first terms, and the second bound of (1) bounds the second terms by $\frac{c}{p} \cdot \frac{16p}{R^2}\|f\|^2$ each. So

$$\sum_{T \in A_e} \|g_T\|^2 \le c\, n^{v-3}\Big(\frac{SR}{p} + \frac{n}{R} + \frac{n}{R^2}\Big)\|f\|^2 + c\Big(\frac{n^{v-3}\Delta b}{p} + \frac{n^{v-2}\tau^2}{pR^2}\Big)\|f\|_\infty^2.$$

For $T \in B_e$, $e \notin T$, so $g_T = \mathcal{T}_T q_T$ with $q_T := \frac{h_T(e)}{R} F_T K_e F_T f$ and $\|q_T\|^2 \le 16p\|f\|^2/R^2$. Put $\eta := K_e F_T f$, so $\|\eta\|^2 \le 4p\|f\|^2$ and $\|\eta\|_\infty \le 2\|f\|_\infty$. For $i \in T$ we have $i \ne e$ and $D_i\eta = K_e D_i F_T f + [D_i, K_e] F_T f$, so $\|D_i\eta\|^2 \le 8p\|D_i F_T f\|^2 + 8p^2\|f\|^2$ by Lemmas 2.2 and 2.3. By (4) applied to η, and $R \ge 1$,

$$\|D_i q_T\|^2 \le \frac{4\|D_i F_T \eta\|^2}{R^2} \le \frac{128p}{R^2}\|D_i F_T f\|^2 + \frac{144p^2}{R^2}\|f\|^2 + \frac{64\tau^2}{R^4}\|f\|_\infty^2.$$

By the first bound of (1) and (6),

$$\sum_{T \in B_e} \|g_T\|^2 \le c\Big(\frac{n^{v-1}p}{R^2} + \frac{n^{v-2}S}{R} + \frac{n^{v-1}p}{R^3}\Big)\|f\|^2 + c\Big(\frac{n^{v-2}\Delta b}{R^2} + \frac{n^{v-1}\tau^2}{pR^4}\Big)\|f\|_\infty^2.$$

Adding both families and dividing by $M \ge cn^v$, the squared error at e is at most $\alpha\|f\|^2 + \beta\|f\|_\infty^2$ with

$$\alpha \le \frac{cS}{n^3}\Big(\frac{R}{p} + \frac{n}{R} + \frac{n^2 p}{R^2}\Big), \qquad \beta \le \frac{c}{n^3}\Big(\frac{n\Delta b}{p} + \frac{n^2\tau^2}{pR^2}\Big) \le c\, n^2 b,$$

using $R \ge 1$, $S \ge 1$, $p \ge 1/n$ (from $R \ge 1$), $\Delta \le vn$ and $\tau^2 \le cn^2 b$. With $R = (np)^{2/3}$, $R/p = n^2 p/R^2 = n^{2/3}p^{-1/3}$ and $n/R \le R/p$, so $\alpha \le cSn^{-7/3}p^{-1/3}$. Since $|J| \le sn^2$, $\sum_e \beta_e \le cn^4 b$. A verifier queries the s edges of the output and accepts if all are present; it always rejects on negative inputs, so $\bar{s} = s$. Theorem 2.1 completes the proof. □

4 2-connected patterns

We use three facts about a 2-connected graph H on $v \ge 3$ vertices.

(F1) For every nonempty $U \subseteq V(H)$ with $|V(H) \setminus U| \ge 2$, at least $|U| + 1$ edges of H have an endpoint in U.

(F2) $H - x$ is connected for every vertex x.

(F3) For distinct edges i, j, every component of the graph $H - \{i, j\}$ obtained by deleting the two edges contains an endpoint of i.

(F2) is the definition. For (F1), let $W := V(H) \setminus U$. Every vertex has degree at least two, so $2e(U) + e(U, W) \ge 2|U|$. Also $e(U, W) \ge 2$, since a single edge between U and W would make its endpoint in W a cut vertex, as $|W| \ge 2$. So $e(U) + e(U, W) \ge |U| + e(U, W)/2 \ge |U| + 1$. For (F3), a component of $H - \{i, j\}$ without an endpoint of i is joined to the rest of H only by j, so j would be a bridge, and a 2-connected graph has none.

Proof of Theorem 1.1. Apply Proposition 3.1 to $G = G_H(n)$ with $\theta := (v-2)/(v-1)$ and $p := c_0 n^{-\theta}$, where $c_0 \in (0,1]$ is a small constant chosen below. Then $np = c_0 n^{1/(v-1)}$, and $R = (np)^{2/3} \geq 1$ for large n. Let $d_0 := \lceil 8\Delta p \rceil$, which is at most $9\Delta p$ since $\Delta p \geq 1$. As in [5, (5.17)], a Chernoff bound under ν, Lemma 2.6 and a union bound show that the maximum degree of a μ^--random input exceeds d_0 with probability at most $b := 2vn\exp(-c\Delta p)$. This is smaller than every power of n.

(P). Let $T \in \mathcal{F}$ and suppose that adding T to a negative input creates a copy $T' \neq T$. Then T' contains an edge of T, so $a := |V(T') \cap V(T)| \geq 2$. If $a \leq v-1$, the edges of T' with an endpoint outside $V(T)$ are not in T and must be present, and by (F1) there are at least $v-a+1$ of them. For each a there are at most cn^{v-a} such T'. If $a = v$, then T' has an edge outside T, and there are at most c such T'. These are increasing events of the outside edges, so by Lemma 2.6,

$$1 - \mu^-(\mathcal{G}_T) \leq c\sum_{a=2}^{v-1} n^{v-a}p^{v-a+1} + cp = c\sum_{a=2}^{v-1} c_0^{v-a+1} n^{(2-a)/(v-1)} + cp \leq c\,c_0 + cp,$$

using $(v-a) - \theta(v-a+1) = (2-a)/(v-1)$. This is at most $1/2$ for small c_0 and large n.

(S). Let $\mathcal{G}_w$ be the event that the present edges outside $\mathrm{st}(w)$ form a graph of maximum degree at most d_0. It contains the event that the whole input has maximum degree at most d_0, so $\mu^-(\mathcal{G}_w^c) \leq b$. Suppose $A_{wa,wa'}(x) = 1$. Then some copy $C \in \mathcal{F}$ contains wa and wa', and all its edges except wa, wa' are present. In particular, all edges of $C - w$ are present, and they lie outside $\mathrm{st}(w)$. By (F2), $C - w$ is connected; fix a spanning tree of it rooted at a. Its $v-2$ non-root vertices can be embedded into the outside graph in at most d_0^{v-2} ways, each vertex being a present neighbour of its parent, and a' is the image of one of them. Accounting for the roles of w, a and a' in H, we get $N_{wa,\mathrm{st}(w)}(x) \leq d_s := v^3 d_0^{v-2}$ on $\mathcal{G}_w$, whatever the assignment inside the star.

(I). Let $\mathcal{G}_I$ be the event that the input has maximum degree at most d_0. Suppose $A_{ij}(x) = 1$ for $i = uu'$, witnessed by a copy C. All edges of $C - \{i,j\}$ are present, and by (F3) every component of $C - \{i,j\}$ contains u or u'. So $C - \{i,j\}$ has a spanning forest with $v-2$ edges and two trees, one containing u and one containing u'; if $C - \{i,j\}$ is connected, delete an edge on the path from u to u' in a spanning tree. Embedding its non-root vertices as above shows that, given the roles of i and j in H, the vertex set of C, and hence j, is one of at most d_0^{v-2} possibilities. So $\#\{j : A_{ij}(x) = 1\} \leq d_m := 2e_H^2 d_0^{v-2}$ on $\mathcal{G}_I$.

Since $d_0 \leq 9vnp$ and $n^{v-2}p^{v-1} = c_0^{v-1}$, both pd_s and pd_m are at most $c\,c_0^{v-1}$. For small c_0 this gives $pd_m \leq 1/5$ and $pd_s \leq 1 \leq R$, so $S \leq 3$. Proposition 3.1 gives

$$\sqrt{\sigma} \leq (t+e_H)\big(c\,n^{-7/3}p^{-1/3} + c\,n^4 b\big)^{1/2} \leq c\,(t+e_H)\,n^{-7/6}p^{-1/6}$$

for large n. So an algorithm with $\sigma \geq 1/3$ needs $t \geq c\,n^{7/6}p^{1/6} = c'\,n^{1+1/(6(v-1))}$ queries, since $\frac{7}{6} - \frac{\theta}{6} = 1 + \frac{1}{6(v-1)}$. For N vertices, take $n = \lfloor N/v \rfloor$ and leave the other vertices isolated. This proves the bound for finding. Since H is connected, finding and detecting have the same query complexity up to a constant factor [4, Lemma 23], as used in [6, Section 2.2]. □

Lemma 4.1. *Let H be a connected graph on v vertices and let B be a 2-connected block of H. Suppose that*

(a) B has the maximum number of edges among the blocks of H, and at most one other block of H is isomorphic to B; or

(b) every component of the graph $H - E(B)$ has fewer than $e(H)/2$ edges.

Write $e_B := |E(B)|$. Then detecting (or finding) B in N-vertex graphs takes at most as many quantum queries as detecting (or finding) H in vN-vertex graphs.

Lemma 4.2. *Let H be connected and B a 2-connected block of H such that*

(c) for every $c \in V(B)$, every component of $H - c$ that avoids $B - c$ has at most $e(H)/2$ edges, counting its edges to c.

Then the lower bound of Theorem 1.1 for B, and every bound from Proposition 3.1 on the blow-up of B, holds for detecting and finding H.

Proof. For $c \in V(B)$ let Br_c be as in the proof of Lemma 4.1. An input G of Proposition 3.1 for B is a subgraph of the blow-up of B; let $h\colon G \to B$ send each vertex to its part. Attach to every $x \in V(G)$ a private copy Γ_x of $\mathrm{Br}_{h(x)}$ with $h(x)$ identified to x, giving G^+ with at most vn vertices, and let $\pi\colon G^+ \to H$ be the homomorphism extending h. A planted copy with each vertex in the part of its role gives a copy of H in G^+. These copies are a constant fraction of $\mathcal{F}$, and Proposition 3.1 bounds t for every constant average success probability σ. Under μ_T the planted copy T is the only copy, so $G - e$ is B-free for $e \in T$; by the claim below every copy of H in G^+ then uses all edges of T, and an algorithm finding H in G^+ finds T. Detection follows by [4, Lemma 23].

Claim: if G has no subgraph isomorphic to B, then G^+ has none isomorphic to H. Let φ be an embedding of H into G^+, $\psi := \pi \circ \varphi \in \mathrm{End}(H)$ and $K := \varphi^{-1}(V(G)) = \psi^{-1}(V(B))$. Then $H[K]$ embeds into G, so it has no subgraph isomorphic to B, and $K \neq \emptyset$ since $|\Gamma_x| < v$. Each component of $H - K$ maps into one component of some $\Gamma_x - x$, whose only neighbour in G is x, so it has exactly one neighbour $k \in K$, with $\varphi(k) = x$. Let T_k be k with its components; ψ is injective on T_k with image in $\mathrm{Br}_{\psi(k)}$, and $k \mapsto \psi(k)$. A block of H with two vertices in K lies in K, by the fan lemma, so $|K \cap V(B)| \leq 1$. A path leaving K returns through the same vertex, so $H[K]$ is connected, and as a union of blocks it has a unique vertex k_0 nearest to B. Root H at B and let $S(k_0)$ be k_0 with everything below it ($S(k_0) = \mathrm{Br}_{k_0}$ if $k_0 \in V(B)$). Then $K \subseteq S(k_0)$, and $H - S(k_0)$ is a component of $H - K$ at k_0. So $H_0 := H - (S(k_0) \setminus \{k_0\})$ embeds into $\mathrm{Br}_{\psi(k_0)}$ with $k_0 \mapsto \psi(k_0)$. Let $b \in V(B)$ be the vertex with $K \subseteq \mathrm{Br}_b$. Since H_0 contains B and every Br_c with $c \neq b$, edge counts give $\psi(k_0) = b$.

If $k_0 \neq b$, then $H_0 - k_0$ is connected and avoids b, so it maps into one component D of $\mathrm{Br}_b - b$. Any component other than the one containing k_0 lies in H_0 together with $B - b$ and k_0, which is too many vertices. So $D \ni k_0$, and with $e(D)$ counting the edges from D to b, $e(H) - e(S(k_0)) = e(H_0) \leq e(D) \leq e(H)/2$ by (c). But $S(k_0)$ lies in D and misses the edges from k_0 towards b, so $e(S(k_0)) < e(D) \leq e(H)/2$, a contradiction.

If $k_0 = b =: u$, then $\psi(u) = u$. Each component C of $H - u$ contains exactly one block Z_C of H at u, and $Z_{C_1} = B$ for the component $C_1 \ni B - u$. Call $C \neq C_1$ free if $C \cap K = \emptyset$; then $C \subseteq T_u$, as is C_1. If $C \neq C_1$ is not free, a path in $H[K]$ from u into C starts with an edge of Z_C, so $Z_C \subseteq K$. For $C = C_1$ or C free, $\psi(C)$ is connected, avoids u and lies in $\mathrm{Br}_u - u$, hence in a component $f(C) \neq C_1$, and $\psi(Z_C) \subseteq Z_{f(C)}$. Put $X_0 := C_1$ and $X_{i+1} := f(X_i)$. Inductively Z_{X_i} contains a copy of B, so X_i is free for $i \geq 1$, as otherwise K would contain a copy of B (if $\mathrm{Br}_u = \{u\}$, then $f(C_1)$ does not exist, which is already a contradiction). Let $a \geq 1$ be minimal with $X_a = X_b$ for some $b > a$. Then $X_{a-1} \neq X_{b-1}$, both lie in T_u and map injectively into X_a with disjoint images, while $|X_a| \leq |X_{b-1}|$ along the chain. So $|X_{a-1}| \leq 0$, a contradiction. □

Proof. Every component of $H - E(B)$ contains exactly one vertex of B; for $c \in V(B)$ let Br_c be the component containing c, rooted at c. Let Γ be the graph obtained from the disjoint union of the Br_c by identifying all roots to one vertex o. The blocks of Γ are the blocks of H other than B, so $e(\Gamma) = e(H) - e_B$. Given an input G, let G^+ be G with a private copy Γ_x of Γ attached at every vertex x, with o identified to x. The new edges are known, and G^+ has at most vN vertices. If G contains a copy of B in which c is mapped to x_c, then mapping Br_c into Γ_{x_c} gives a copy of H in G^+.

Conversely, let φ be an embedding of H into G^+, and suppose G contains no copy of B. Every block of G^+ lies in G or in a single gadget, since a gadget meets the rest of G^+ in one vertex. The image of a 2-connected subgraph lies in one block of G^+. In particular, $\varphi(B)$ lies in a block of some gadget Γ_x. If $x \notin \varphi(V(H))$, then all of $\varphi(H)$ lies in $\Gamma_x - x$, since H is connected, which is impossible as $e(\Gamma) < e(H)$. So $x = \varphi(u)$ for a vertex u of H. Let K be the component of $H - u$ that contains $B - u$, and $Q := H[K \cup \{u\}]$. Since $\varphi(K)$ is connected and avoids x, it lies in $\Gamma_x - x$, hence in one copy of $\mathrm{Br}_c - c$, and φ embeds Q into Br_c with $u \mapsto c$. Also, a shortest path in H from u to a vertex of K stays in Q, so $\mathrm{dist}_{\mathrm{Br}_c}(c, \varphi(y)) \le \mathrm{dist}_H(u, y)$ for $y \in K$.

Let $e \in V(B)$ be the vertex with $u \in \mathrm{Br}_e$. Then Q contains $Q_B := H - (V(\mathrm{Br}_e) \setminus \{e\})$, which consists of B and all $\mathrm{Br}_{c'}$ with $c' \neq e$. Indeed, if $u \neq e$, this subgraph is connected, contains B and avoids u; if $u = e$, then $Q_B - e$ is connected, since $B - e$ is connected and every other branch attaches at a vertex of $B - e$.

(b) If $c \neq e$, then $\mathrm{Br}_c \subseteq Q_B - E(B)$, so $e(\mathrm{Br}_c) < e(Q)$, which contradicts the embedding. If $c = e$, then $e(H) - e(\mathrm{Br}_e) = e(Q_B) \le e(Q) \le e(\mathrm{Br}_e)$, so Br_e has at least $e(H)/2$ edges, contrary to (b).

(a) Call a block of H marked if it is isomorphic to B. The image of a marked block lies in a block of G^+ that contains a copy of B. Not in G, so in a gadget, where every block is isomorphic to a block of H other than B and has at most e_B edges. So the image is a whole marked block of a gadget, and different marked blocks of H have edge-disjoint images, hence occupy different marked blocks. A gadget contains one marked block fewer than H. If H has only one marked block, this is already a contradiction. Otherwise let Y be the other one, and Y° its copy in Γ. Both B and Y map to copies of Y°, in different gadgets Γ_x and $\Gamma_{x'}$, where $x = \varphi(u)$ as above. Then $u \notin V(Y)$, since the image of Y lies in $\Gamma_{x'}$, whose only vertex in G is $x' \neq x$. Moreover $V(Y) \cap K = \emptyset$: otherwise the connected graph Y would lie in K, and its image in $\Gamma_x - x$. Let $D := \mathrm{dist}_H(V(B), V(Y))$. Since $\varphi(B - u)$ lies in the copy of Y° in Γ_x, the branch Br_c above is the one containing Y, and $D = \mathrm{dist}_{\mathrm{Br}_c}(c, V(Y))$. If $u \in V(B)$, then x lies in the image of B, so $D = 0$ and B and Y share a vertex; it is not in K, so it is u, which contradicts $u \notin V(Y)$. Otherwise $B \subseteq K$, and the distance bound above gives $\mathrm{dist}_H(u, V(B)) \ge D$. But a shortest path from $V(B)$ to $V(Y)$ has length D, starts in K and ends outside $K \cup \{u\}$, so it passes through u after at most $D - 1$ steps, and $\mathrm{dist}_H(u, V(B)) \le D - 1$, a contradiction.

In both cases G contains a copy of B whenever G^+ contains a copy of H. For finding, the proof shows that in a copy of H in G^+ some block isomorphic to B lies in G, and the gadget vertices are known. □

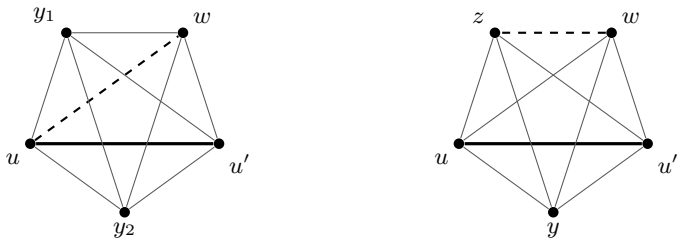
Corollary 1.2 follows from Lemma 4.1 and Theorem 1.1 applied to B.

5 Cliques

Fix $4 \le k \le 7$, $H = K_k$, $s := \binom{k}{2}$, $\theta := 2/(k+1)$ and $p := c_0 n^{-\theta}$. Now $G = G_{K_k}(n)$ is the complete k-partite graph with parts of size n, every member of $\mathcal{F}$ has one vertex in each part, $\Delta = (k-1)n$ and $M = n^k$.

Lemma 5.1. *There are $C_2, \delta > 0$ depending only on k such that, for $n \ge n_0(k, c_0)$, outside an event of μ^--probability at most $\exp(-n^\delta)$, every edge i satisfies $\#\{j \neq i : A_{ij}(x) = 1\} \le C_2 c_0^{s-2} n^\theta$.*

Proof. Fix $i = uu'$. If $A_{ij}(x) = 1$, then a copy of K_k on a vertex set $U \ni u, u'$ contains j and has all its edges except i, j present. Either $j = uw$ or $j = u'w$ shares a vertex with i, or $j = wz$ is disjoint from i (Figure 1). So the number of such j is at most $X_u + X_{u'} + X_0$, where X_u counts the sets U



(a) $j = uw$ shares the vertex u with i (b) $j = wz$ is disjoint from i

Figure 1: The two patterns in the proof of Lemma 5.1, drawn for $k = 5$. The thick edge is the queried coordinate $i = uu'$, the dashed edge the interacting coordinate j, and the thin edges must be present. Pattern (a) has $k-3$ further vertices, pattern (b) has $k-4$.

with a distinguished vertex w such that all pairs in U except uu', uw are present, $X_{u'}$ is symmetric, and X_0 counts the sets U with a distinguished pair $\{w, z\}$ such that all pairs except uu', wz are present.

Under ν the variables $(x_f)_{f\neq i}$ are independent Booleans, and each count is a homogeneous polynomial of degree $q := s-2$ in them with coefficients in $\{0,1\}$, since a configuration is determined by its set of required edges (here $k \geq 4$). Under ν, each has expectation at most $\bar{\mu} := k^2 n^{k-2} p^{s-2} = k^2 c_0^{s-2} n^{\theta}$, using $\theta(s-2) = k-2-\theta$. We use the inequality of Kim and Vu [7]: for a polynomial X of degree at most q with coefficients in $[0,1]$ in m independent Boolean variables and $\lambda > 1$,

$$\Pr\big[|X - \mathbb{E}X| > a_q (\mathcal{E}_0 \mathcal{E}_1)^{1/2} \lambda^q\big] \leq 2e^2 e^{-\lambda} m^{q-1},$$

where $a_q = 8^q (q!)^{1/2}$, $\partial_A X$ is the partial derivative of X with respect to the variables in a set A, $\mathcal{E}_0 := \max_{|A|\geq 0} \mathbb{E}[\partial_A X]$ and $\mathcal{E}_1 := \max_{|A|\geq 1} \mathbb{E}[\partial_A X]$.

Let $X = X_u$ or X_0 and let A be a nonempty set of variables with $\partial_A X \neq 0$. Every required edge has an endpoint outside $\{u, u'\}$. Let Σ be the set of these endpoints of the edges of A, and fix the roles that the vertices of Σ play in the pattern, at a cost of a factor k^k. Let $e(\Sigma)$ be the number of required edges inside $\Sigma \cup \{u, u'\}$, so $|A| \leq e(\Sigma)$. A configuration counted by $\partial_A X$ is determined by its other $k-2-|\Sigma|$ vertices, and it needs its other $s-2-|A| \geq s-2-e(\Sigma)$ edges. Hence $\mathbb{E}[\partial_A X] \leq k^k n^{k-2-|\Sigma|} p^{s-2-e(\Sigma)} = k^{k-2} \bar{\mu}/\lambda_\Sigma$ with $\lambda_\Sigma := n^{|\Sigma|} p^{e(\Sigma)}$. We claim that

$$\lambda_\Sigma \geq c_0^s \, n^{1/(k+1)} \qquad \text{for every nonempty } \Sigma. \tag{7}$$

For X_u, let Σ contain a of the vertices other than u, u', w, and $\epsilon \in \{0,1\}$ copies of w. Then $e(\Sigma) = 2a + \binom{a}{2} + \epsilon(1+a)$, and $\log_n(\lambda_\Sigma / c_0^{e(\Sigma)})$ equals $a(k-2-a)/(k+1)$ for $\epsilon = 0$ and $1 \leq a \leq k-3$, and $(-a^2 + a(k-4) + k - 1)/(k+1)$ for $\epsilon = 1$ and $0 \leq a \leq k-3$. The first is at least $(k-3)/(k+1)$. The second is concave in a, with values $(k-1)/(k+1)$ at $a = 0$ and $2/(k+1)$ at $a = k-3$. For X_0, with a vertices other than u, u', w, z and $\epsilon \in \{0,1,2\}$ of w, z, $e(\Sigma) = 2a + \binom{a}{2} + \epsilon(2+a)$. The exponents are $a(k-2-a)/(k+1)$, $(-a^2+a(k-4)+k-3)/(k+1)$ and $(-a^2+a(k-6)+2k-6)/(k+1)$, which are at least $(k-3)/(k+1)$, $(k-3)/(k+1)$ and $2/(k+1)$ for $0 \leq a \leq k-4$ (and $a \geq 1$ when $\epsilon = 0$). All are at least $1/(k+1)$, and $c_0^{e(\Sigma)} \geq c_0^s$, which proves (7).

So $\mathcal{E}_1 \leq k^{k-2} c_0^{-s} n^{-1/(k+1)} \bar{\mu}$, and $\mathcal{E}_0 \leq \bar{\mu}$ for $n \geq n_0(k, c_0)$. With $\lambda := n^{1/(2(k+1)q)}/C_3$ for a suitable $C_3 = C_3(k, c_0)$, $a_q(\mathcal{E}_0\mathcal{E}_1)^{1/2}\lambda^q \leq \bar{\mu}$, so $X \leq 2\bar{\mu}$ outside probability $2e^2 e^{-\lambda} m^{q-1}$, where $m \leq sn^2$; here $\lambda > 1$ for large n. A union bound over the sn^2 edges i and the three counts, and Lemma 2.6 (the bad event is increasing), prove the lemma with $C_2 = 6k^2$ and any $\delta < 1/(2(k+1)q)$, for large n. □

Proof of Theorem 1.3. We verify the hypotheses of Proposition 3.1. Let $d_0 := \lceil 8\Delta p \rceil \le 9\Delta p$, and let b be $\exp(-n^\delta)$ plus the probability $2kn\exp(-c\Delta p)$ that the maximum degree exceeds d_0.

(P). A copy $T' \neq T$ created by adding T shares a vertices with T for some $2 \le a \le k-1$ and needs its $s - \binom{a}{2}$ edges outside T to be present. There are at most $\binom{k}{a} n^{k-a}$ such copies. Since $s - \binom{a}{2} = (k-a)(k+a-1)/2$, the expected number is at most $\binom{k}{a} c_0^{s-\binom{a}{2}} n^{(k-a)(2-a)/(k+1)} \le \binom{k}{a} c_0^{s-\binom{a}{2}}$, and Lemma 2.6 gives $\mu^-(\mathcal{G}_T) \ge 1/2$ for small c_0.

(S). Fix the edges outside $\mathrm{st}(w)$. A copy of K_k through w consists of star edges $wa_1, \dots, wa_{k-1}$ and a K_{k-1} on $a_1, \dots, a_{k-1}$. If $A_{wa,wa'}(x) = 1$, then aa' is a present edge outside the star. So $N_{wa,\mathrm{st}(w)}(x)$ is at most the degree of a outside the star. On the event $\mathcal{G}_w$ that the outside graph has maximum degree at most d_0, (S) holds with $d_s := d_0 \le 9(k-1)np$. Then

$$\frac{pd_s}{R} \le \frac{9(k-1)np^2}{(np)^{2/3}} = 9(k-1)c_0^{4/3} n^{\frac{1}{3}-\frac{4}{3}\theta} = 9(k-1)c_0^{4/3} n^{\frac{k-7}{3(k+1)}} \le 9(k-1),$$

so $S \le 2 + 81(k-1)^2$ for $k \le 7$.

(I) holds with $d_m := C_2 c_0^{s-2} n^\theta$ by Lemma 5.1, and $pd_m = C_2 c_0^{s-1} \le 1/5$ for small c_0.

Proposition 3.1 gives $\sqrt{\sigma} \le c(t+s)n^{-7/6}p^{-1/6}$ for large n, and $n^{7/6}p^{1/6} = c_0^{1/6} n^{7/6 - 1/(3(k+1))}$. As before, take $n = \lfloor N/k \rfloor$ and use [4, Lemma 23] for detection. □

Proof of Corollary 1.4. By [6, Theorem 1.3(i)], $Q(\mathrm{Sub}_H) = \Omega(Q(\mathrm{Sub}_{K_r}))$ with $r = \min\{\chi, 6\}$, and the same holds for finding. Apply Theorem 1.3 with $r = 4, 5, 6$. □

6 Denser patterns: extension counts

The degree bounds of Section 4 count the extensions of a spanning tree, and they hold up to $p \approx n^{-(v-2)/(v-1)}$. For denser H more edges of a copy must be present, and counting the extensions of a larger pattern allows larger densities. This section turns this into a general criterion and applies it to complete bipartite graphs.

Rooted patterns. A *rooted pattern* in H is a triple (S, Q, F) with $Q \subseteq S \subseteq V(H)$ and $F \subseteq E(H[S])$ such that no edge of F lies inside Q. For $0 < \theta < 1$ put

$$\varepsilon_\theta(S, Q, F) := \max_{Q \subseteq U \subseteq S} \big(|S \setminus U| - \theta |F \setminus F[U]| \big) \ge 0,$$

where $F[U]$ is the set of edges of F inside U; the term $U = S$ is 0. The pattern is *safe at* θ if $|U \setminus Q| - \theta|F[U]| > 0$ for every U with $Q \subsetneq U \subseteq S$. For an injective map ρ from Q to $V(G)$ and a set $Z \subseteq V(G)$, let $X_\rho(x)$ be the number of injective maps $\psi : S \to V(G) \setminus Z$ that extend ρ and map every edge of F to an edge of G present in x.

Lemma 6.1. *Let (S, Q, F) be a rooted pattern in which every vertex of $S \setminus Q$ lies on an edge of F, let $0 < \theta < 1$, $p = c_0 n^{-\theta}$ with $c_0 \le 1$, and $\varepsilon := \varepsilon_\theta(S, Q, F)$. There is C depending only on H, not on c_0, such that for $n \ge n_0(H)$, every ρ and Z,*

$$\nu\big[X_\rho > Cn^\varepsilon (\log n)^{|F|}\big] \le n^{-20}.$$

If the pattern is safe at θ, then there is $\delta > 0$ such that $\nu[X_\rho > C\bar{\mu}] \le \exp(-n^\delta)$ for $n \ge n_0(H, c_0)$, with the same C, where $\bar{\mu} := (vn)^{|S \setminus Q|} p^{|F|}$.

Proof. Put $q := |F|$; if $q = 0$ then $S = Q$ and $X_\rho \le 1$. Each ψ contributes the monomial $\prod_{f\in F} x_{\psi(f)}$, of degree q since ψ is injective. The monomial determines the image of $S \setminus Q$, because every vertex of $S \setminus Q$ lies on an edge of F, so at most $v!$ maps ψ give the same monomial. Hence $X_\rho/v!$ is a homogeneous polynomial of degree q with coefficients in $[0,1]$ in the independent Boolean variables x_g, $g \in J$. Let A be a set of variables with $\partial_A X_\rho \neq 0$, and let $U \supseteq Q$ consist of Q and the vertices of S that ψ maps to endpoints of edges of A, for some ψ counted by $\partial_A X_\rho$. Then $|A| \le |F[U]|$. There are at most 2^v choices of U and at most $(2e_H + v)^v$ of $\psi|_U$, whose values are endpoints of edges of A or values of ρ; the other vertices have at most vn images each, and the other $q - |A| \ge q - |F[U]|$ edges must be present. So

$$\mathbb{E}[\partial_A X_\rho] \le v!\, 2^v (2e_H + v)^v \max_{Q\subseteq U\subseteq S} (vn)^{|S\setminus U|} p^{q-|F[U]|} \le C' n^{\varepsilon},$$

including $A = \emptyset$. The inequality of Kim and Vu, quoted in the proof of Lemma 5.1, with $\lambda := 22q \log n$ gives $X_\rho \le C' n^\varepsilon + v!\, a_q C' n^\varepsilon \lambda^q$ outside probability $2e^2 e^{-\lambda}(e_H n^2)^{q-1} \le n^{-20}$ for large n.

If the pattern is safe, let $\delta_0 > 0$ be the minimum of $|U \setminus Q| - \theta|F[U]|$ over $U \neq Q$. The term for $U \neq Q$ above equals $v^{|S\setminus U|}\bar\mu\, v^{-|S\setminus Q|} c_0^{-|F[U]|} n^{-(|U\setminus Q| - \theta|F[U]|)}$, so $\mathcal{E}_1 \le C'' c_0^{-q} n^{-\delta_0}\bar\mu$ and $\mathcal{E}_0 \le C''\bar\mu$ for large n. With $\lambda := n^{\delta_0/(2q)}/C_3$ as in the proof of Lemma 5.1, the deviation is at most $C''\bar\mu$ outside probability $\exp(-n^\delta)$ for every $\delta < \delta_0/(2q)$ and large n. Since $p \le n^{-\theta}$, the constants C' and C'' depend only on H. □

Let $e_{\max}(j)$ be the largest number of edges of H spanned by j vertices, and

$$\theta_P(H) := \max_{2\le j\le v-1} \frac{v - j}{e_H - e_{\max}(j)}.$$

We call θ *star-admissible* if for every vertex x_0 of H and all distinct neighbours y, z of x_0 there is a set $S \subseteq V(H) \setminus \{x_0\}$ containing y and z such that the pattern $(S, \{y\}, E(H[S]))$ satisfies $\varepsilon_\theta < 2/3 + \theta/3$, or $\varepsilon_\theta = 2/3 + \theta/3$ and is safe at θ.

Theorem 6.2. *Let H be a fixed connected graph on $v \ge 3$ vertices and let $\theta_P(H) \le \theta < 1$ be star-admissible. If $\theta = \theta_P(H)$, assume in addition that $(V(H), i, E(H) \setminus \{i, j\})$ is safe at θ for all distinct edges i, j of H. Then detecting or finding H takes $\Omega(N^{7/6-\theta/6})$ quantum queries.*

Proof. Let $p := c_0 n^{-\theta}$ and verify the hypotheses of Proposition 3.1; then $R = (np)^{2/3} \ge 1$ for large n. Let $\eta := n^{-15}$ be the failure probability b of Proposition 3.1.

(P). A copy $T' \neq T$ created by adding T shares $j \ge 2$ vertices with T. For $j \le v - 1$ there are at most $\binom{v}{j} C_H n^{v-j}$ such copies, each needing at least $e_H - e_{\max}(j)$ present edges outside T, so the expected number is at most $\binom{v}{j} C_H c_0 n^{(v-j)-\theta(e_H - e_{\max}(j))} \le \binom{v}{j} C_H c_0$. For $j = v$ the expected number is $O(p)$. Lemma 2.6 gives $\mu^-(\mathcal{G}_T) \ge 1/2$ for small c_0.

(S). Fix w. Suppose $A_{wu,wu'}(x) = 1$ through a copy $\varphi(H)$ with $\varphi(x_0) = w$, $\varphi(y) = u$, $\varphi(z) = u'$, and let S be the set chosen for (x_0, y, z). If some vertex of $S \setminus \{y\}$ lies on no edge of $H[S]$, then the term for $U = S$ minus that vertex gives $\varepsilon_\theta \ge 1 > 2/3 + \theta/3$, so this does not happen. The edges of $H[S]$ avoid x_0, so φ maps them outside $\mathrm{st}(w)$, where they are present. Hence $\varphi|_S$ is counted by X_ρ with $\rho(y) = u$ and $Z = \{w\}$, and $u' = \varphi(z)$. So $N_{wu,\mathrm{st}(w)}(x)$ is at most the sum of these counts over the at most v^3 triples (x_0, y, z). They depend only on the edges outside $\mathrm{st}(w)$, and the event that one of them exceeds the bound of Lemma 6.1 is increasing. By Lemma 6.1, the union bound over u and Lemma 2.6, (S) holds with failure probability at most η and either $d_s := Cv^3 n^\varepsilon (\log n)^{e_H}$ with $\varepsilon < 2/3 + \theta/3$, or $d_s := Cv^3\bar\mu$ with $\bar\mu \le c\, c_0^{|F|} n^{2/3+\theta/3}$. In the first case $pd_s/R = c_0^{1/3} n^{\varepsilon - 2/3 - \theta/3} Cv^3 (\log n)^{e_H} \to 0$; in the second $pd_s/R \le c c_0^{1/3}$. (Different triples may use different cases; the bounds add.) So $pd_s \le R$ for small c_0 and large n.

(I). Suppose $A_{ij}(x) = 1$ through a copy $\varphi(H)$, and let i_H, j_H be the edges of H mapped to i and j. Then φ is counted by X_ρ for the pattern $P_{i_H,j_H} := (V(H), i_H, E(H) \setminus \{i_H, j_H\})$ with $\rho = \varphi|_{i_H}$ and $Z = \emptyset$, and $j = \varphi(j_H)$. Since $\theta_P \le \theta < 1$, H has minimum degree at least 2: a vertex of degree one gives the ratio 1 for $a = v - 1$. So every vertex outside i_H lies on an edge of this pattern. For $\overline{U} := V(H)\setminus U$, the number of edges of P_{i_H,j_H} that meet $\overline{U}$ is at least $e'-1$, where e' is the number of edges of H that meet $\overline{U}$. Since U contains i_H, it has at least two vertices and $e_H - e' \le e_{\max}(v - |\overline{U}|)$, so $|\overline{U}| \le \theta_P e' \le \theta e'$ for $\overline{U} \ne \emptyset$. Hence each term of $\varepsilon_\theta(P_{i_H,j_H})$ is at most $|\overline{U}| - \theta(e'-1) \le \theta$, with strict inequality when $\theta > \theta_P$. If $\theta > \theta_P$, Lemma 6.1 gives $d_m := Ce_H^2 n^\varepsilon (\log n)^{e_H}$ with $\varepsilon < \theta$, so $pd_m \to 0$. If $\theta = \theta_P$, safety gives $d_m := Ce_H^2\bar{\mu}$ with $p\bar{\mu} = c\, c_0^{e_H-1} n^{(v-2)-\theta(e_H-1)} \le c c_0^{e_H-1}$, since the term for $U = i_H$ is at most θ. So $pd_m \le 1/5$ for small c_0. The failure probability is at most η by Lemma 2.6.

Proposition 3.1 gives $\sqrt{\sigma} \le c(t + e_H) n^{-7/6} p^{-1/6}$ for large n, since $n^4\eta \le n^{-11}$. Take $n = \lfloor N/v \rfloor$ and use [4, Lemma 23] for detection. □

Corollary 6.3. *Let $3 \le a \le b$ and $\theta_{a,b} := (3a + 3b - 8)/(3(a-1)b + 1)$. Detecting or finding $K_{a,b}$ takes $\Omega(N^{7/6-\theta_{a,b}/6})$ quantum queries. Detecting or finding $K_{2,3}$ takes $\Omega(N^{16/15})$, and $K_{2,b}$ with $b \ge 4$ takes $\Omega(N^{15/14})$.*

For $K_{r,r}$ with $r \ge 3$ the exponent is $7/6 - (3r-4)/(3(3r^2 - 3r + 1))$: 41/38 for $r = 3$, 81/74 for $r = 4$ and 615/542 ≈ 1.135 for $r = 10$, where Theorem 1.1 gives 115/114 and [6] gives 181/180. A comparison with exact fractions shows that it exceeds β_r of [6, (68)] exactly for $r \le 18$ among $10 \le r \le 20$; since β_r increases and the exponent here stays below 7/6, while $\beta_{20} > 7/6$, the bounds of [6] are larger for all $r \ge 19$.

Proof. Let $H = K_{a,b}$ with sides A and B, $|A| = a \le b = |B|$. A copy sharing $a_1 \ge 1$ vertices of A and $b_1 \ge 1$ of B with T, $(a_1, b_1) \ne (a, b)$, gives the ratio $(a + b - a_1 - b_1)/(ab - a_1 b_1)$. With $\bar{a} = a - a_1$ and $\bar{b} = b - b_1$, the quantity $(a+b-2)(\bar{a}b + \bar{b}a - \bar{a}\bar{b}) - (ab-1)(\bar{a}+\bar{b})$ is bilinear in $(\bar{a}, \bar{b})$, with values 0, $(a-1)(b-1)^2$, $(b-1)(a-1)^2$ and 0 at the corners of $[0, a-1] \times [0, b-1]$, so it is nonnegative and $\theta_P = (a + b - 2)/(ab - 1)$; shared sets with $a_1 = 0$ or $b_1 = 0$ contain no edge of T and cannot create a copy.

Let $a \ge 3$ and $\theta = \theta_{a,b}$. First, $\theta > \theta_P$, since

$$(3a + 3b - 8)(ab - 1) - (a + b - 2)(3ab - 3b + 1) = 3b^2 + ab - 4a - 10b + 10 > 0$$

for $3 \le a \le b$. Also $\theta < 1$, since $3(a-1)b + 1 - (3a + 3b - 8) = 3((a-2)(b-1) + 1) > 0$, and $\theta(a-1) \ge 1$, which is equivalent to $(3a - 8)(a - 1) \ge 1$. For star-admissibility take $S = V(H) \setminus \{x_0\}$. If $x_0 \in A$, then $y, z \in B$, and the set $\overline{U} = S \setminus U$ with α vertices in A and β in B meets $\alpha b + \beta(a-1) - \alpha\beta$ edges of $H - x_0$. The term $|\overline{U}| - \theta(\alpha b + \beta(a-1) - \alpha\beta)$ is bilinear in $(\alpha, \beta) \in [0, a-1] \times [0, b-1]$, so its maximum is at a corner. The corner values are 0, $(a-1)(1-\theta b) \le 0$, $(b-1)(1 - \theta(a-1)) \le 0$ and $a + b - 2 - \theta(a-1)b = 2/3 + \theta/3$, by the definition of θ. So $\varepsilon_\theta = 2/3 + \theta/3$. The pattern is safe: for $U \ne \{y\}$ with α' vertices of A and β' vertices of $B \setminus \{y\}$, $|U \setminus \{y\}| - \theta|F[U]| = \alpha' + \beta' - \theta\alpha'(\beta' + 1)$. For $\alpha' = 0$ this is $\beta' > 0$. For $\alpha' \ge 1$ it is linear in β', with value $\alpha'(1-\theta) > 0$ at $\beta' = 0$ and $\alpha'(1 - \theta b) + b - 1$ at $\beta' = b - 1$, which is linear in α' with values $b(1-\theta) > 0$ and $2/3 + \theta/3 > 0$ at $\alpha' = 1$ and $\alpha' = a - 1$. If $x_0 \in B$, the same computation with $\alpha \le a - 1$ vertices in A and $\beta \le b - 1$ in B gives the corners 0, $(a-1)(1 - \theta(b-1)) \le 0$, $(b-1)(1 - \theta a) \le 0$ and $a + b - 2 - \theta a(b-1) \le a + b - 2 - \theta(a-1)b$; if $a < b$ the last is strictly smaller, and if $a = b$ the case is symmetric to $x_0 \in A$. Theorem 6.2 applies.

Let $a = 2$, and let x_0, y, z be as in the definition, so y, z lie on the side not containing x_0. Take $S = \{y, l, z\}$ with $l \ne x_0$ on the side of x_0, so that $H[S]$ is the path $y\, l\, z$. The terms of ε_θ are

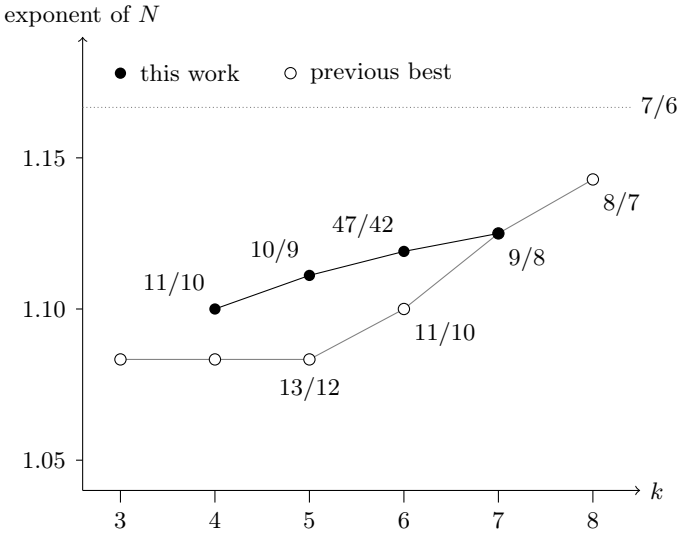


Figure 2: Lower bound exponents for K_k detection. Open circles: the previous best, from [5] for $k \le 5$ and from the clique collision bound of [6, Corollary 4.6] for $k \ge 6$, which is $1/2 + \ell(2k-\ell-1)/(2(k-1)(\ell+1))$ maximized over ℓ, minus $o(1)$. Filled circles: Theorem 1.3.

$2-2\theta$, $1-\theta$, $1-2\theta$ and 0 for $U = \{y\}$, $\{y,l\}$, $\{y,z\}$, S, so $\varepsilon_\theta = 2-2\theta \le 2/3+\theta/3$ if and only if $\theta \ge 4/7$, and the path is safe since $1-\theta$, 1 and $2-2\theta$ are positive. Here $\theta_P = b/(2b-1)$, which is $3/5$ for $b=3$, $4/7$ for $b=4$ and smaller for $b \ge 5$. So $\theta = \max\{4/7, b/(2b-1)\}$ is star-admissible. If $b \in \{3,4\}$, then $\theta = \theta_P$, and the patterns $P_{i,j}$ are safe: with $i = uu'$, $u \in A$, a set $U \setminus i$ with $\alpha \in \{0,1\}$ vertices of A and β vertices of B spans at most $\beta + \alpha(1+\beta)$ edges of $P_{i,j}$, and at most $2b-2$ if $U = V(H)$. For $\alpha = 0$ the safety term is $\beta(1-\theta) > 0$; for $\alpha = 1$ and $\beta \le b-2$ it is at least $1+\beta-\theta(1+2\beta) > 0$, since $(1+\beta)/(1+2\beta) \ge (b-1)/(2b-3) > b/(2b-1)$; and for $U = V(H)$ it is $b-\theta(2b-2) > 0$. Theorem 6.2 gives the exponents $7/6-1/10 = 16/15$ and $7/6-2/21 = 15/14$. □

Remark 6.4. The octahedron $K_{2,2,2}$ has chromatic number 3, so the best previous bound known to us is $N^{13/12}$ [5]. Here $\theta_P = 4/11$. For a vertex x_0 and adjacent neighbours y, z, the pattern $S = \{y,z\}$ has $\varepsilon_\theta = 1-\theta$. For the non-adjacent pairs, $S = V(H) \setminus \{x_0\}$ induces a wheel on five vertices with hub the antipode of x_0, and $\varepsilon_\theta = 4-8\theta$ for $\theta \le 1/2$, which equals $2/3+\theta/3$ at $\theta = 2/5$; this pattern is safe at $2/5$. Theorem 6.2 gives $\Omega(N^{11/10})$.

The conditions of Theorem 6.2 are finite and can be checked by a computer. A script computes the smallest $\theta \ge \theta_P$ that is star-admissible and, for $\theta > \theta_P$ automatically and for $\theta = \theta_P$ through the safety of $P_{i,j}$, satisfies (I), minimizing over all sets S for each triple (x_0, y, z); it also checks that each condition that is tight at this θ has a safe pattern. It reproduces Theorems 1.1 and 1.3 for C_3, C_4, K_4 and K_5, and it gives the value $\theta_{a,b}$ of Corollary 6.3 (and $\max\{4/7, b/(2b-1)\}$ for $a=2$) by brute force for $a+b \le 8$ and by a computation over vertex types for all $2 \le a \le b \le 30$. The scripts are in `https://github.com/ssenge/LAIF`, in the folders `complete-bipartite-quantum-lower-bounds` and `two-connected-subgraph-quantum-lower-bounds`.

7 Limits of the method

Whenever its hypotheses hold with bounded pd_s/R, Proposition 3.1 gives $t = \Omega(n^{7/6}p^{1/6})$, so the bound is limited by the largest density at which (P), (S) and (I) hold.

Cut vertices. Proposition 3.1 with the degree bounds of Section 4 does not apply directly when a cut vertex w of H joins two blocks with cycles. In the star of a vertex playing the role of w, an edge wa in one block interacts with every edge wa' in the other block for which a' lies in a present copy of that block minus its edges at w. When this graph is a tree, as for two four-cycles sharing a vertex, the number of such a' is of order $n(np)^{\beta-2}$ for a block with β vertices, so $pd_s \approx (np)^{\beta-1}$ exceeds R at every density with $np \to \infty$. Lemma 4.1 avoids the problem by reducing to a single block. For denser blocks Theorem 6.2 applies directly, since a dense pattern on the far side of the cut vertex has few extensions. For two copies of $K_{3,3}$ sharing a vertex, $\theta_P = 5/9$ and $\theta = 13/19$ is star-admissible, with the pattern $K_{3,3}$ minus the cut vertex on the far side, which gives $\Omega(N^{20/19})$. Lemma 4.1 transfers every lower bound for B to H, so with Corollary 6.3 it gives more here: $\Omega(N^{41/38})$, by (a).

Densities. The planting condition alone allows $p \approx n^{-\theta_P}$, with θ_P as in Section 6. For cycles $\theta_P = (v-2)/(v-1)$, so for them hypothesis (P) allows no larger density. For K_k with $k \leq 7$ and for $K_{2,3}$ and $K_{2,4}$ the bounds reach θ_P. For $K_{r,r}$ with $r \geq 3$ star-admissibility, minimized over all sets S, is the binding condition, and for K_k with $k \geq 8$ it caps the density at $n^{-1/4}$, which gives only $N^{9/8}$. The exponent $7/6 - \theta/6$ stays below $7/6$ for every H, so for large r the random constructions of [6] give more for $K_{r,r}$.

Large cliques. For $k \geq 8$, the planting condition allows p above $n^{-1/4}$, but there $pd_s \approx np^2$ exceeds $R = (np)^{2/3}$, and $S \approx n^2p^4/R^2$. The resulting heuristic error bound is of order $n^{-3}(R/p + n^2p^3/R + n^2p/R^2)$. For $p \geq n^{-1/4}$ it is smallest at $R = np^2$, where it is of order p/n^2, which gives $t \gtrsim np^{-1/2}$, decreasing in p. So the best choice is $p \approx n^{-1/4}$, giving $N^{9/8}$, below the bound $N^{8/7-o(1)}$ of [6] for K_8 (Figure 2). Lemma 3.2 may be lossy here, so this is a limit of the present analysis rather than of conditioned random graphs.

Tightness. For triangles, the bound $n^{7/6}p^{1/6}$ gives lower bounds for sparse graphs that are tight up to polylogarithmic factors in a range of edge counts [5]. We do not know whether $n^{7/6}p^{1/6}$ is tight on our hard inputs for any other pattern.

Use of AI

Claude Opus 5.5 (Anthropic) was used throughout this research: to explore approaches, to develop and check proofs, to write the check scripts, and to draft the text.

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